

HOSPITAL MANAGEMENT SYSTEM

A hospital management system is an information system that manages the aspects of a hospital. This may include the administrative, financial, and medical processing. It is an integrated end-to-end Hospital Management System that provides relevant information across the hospital to support effective decision-making for patient care, hospital administration, and critical financial accounting, in a seamless flow. This program can look after inpatients, OPD patients, records, database treatment, status illness, billing, etc. It also maintains their hospital info such as ward ID, doctor in charge, department administering, etc.

NEED OF HOSPITAL MANAGEMENT SYSTEM

1. **Efficient Patient Management:** Streamlines the process of registering and tracking patient information.
2. **Quick Access to Records:** Allows easy retrieval of patient and doctor details, speeding up service delivery.
3. **Accurate Billing:** Automates billing processes to minimize errors and improve financial management.
4. **Appointment Scheduling:** Simplifies appointment management, reducing wait times and enhancing patient satisfaction.
5. **Data Organization:** Keeps hospital data organized, improving overall efficiency.
6. **Resource Optimization:** Helps allocate doctors and resources effectively based on need and specialization.
7. **Improved Decision-Making:** Provides quick access to critical data, aiding in timely and informed decisions.

8. Error Reduction: Minimizes manual errors in patient records and administrative tasks.
9. Regulatory Compliance: Ensures data is managed securely and in compliance with healthcare standards.
10. Enhanced Communication: Facilitates better coordination among hospital departments.

#PROJECT - PROGRAM FOR MANAGEMENT OF HOSPITAL AND DOCTORS AND PATIENTS DETAILS#.

#Database connection

```
import mysql.connector
def connect_to_db():
    return mysql.connector.connect(
        host="localhost",
        user="root",
        password="mansi",
        database="HOSPITAL_MANAGEMENT"
```

#program for add patients

```
def add_patient(cursor):
    name=input("Enter patient's name:")
    age=int(input("Enter patient's age:"))
    gender=input("Enter patient's gender:")
    phone=int(input("Enter phone number:"))
    admission_date=input("Enter admission date (YYYY-MM-DD):")
    query = "INSERT INTO PATIENTS (Name, Age, Gender,
Phone,Admission_date) VALUES (%s, %s, %s, %s,%s)"
    cursor.execute(query, (name, age, gender, phone,admission_date))
    print("Patient added successfully!")
```

#program for view patients

```
def view_patients(cursor):
    cursor.execute("select * from PATIENTS")
    records=cursor.fetchall()
    print("Patients records:")
    for record in records:
        print(record)
```

#program for update patients

```
def update_patients(cursor):
    patient_id = int(input("Enter the Patient ID to update: "))
    print("1. Update Name")
    print("2. Update Age")
    print("3. Update Gender")
    print("4. Update Phone Number")
    choice = int(input("Choose the detail to update: "))
    if choice == 1:
        new_name = input("Enter new name: ")
        query = "UPDATE PATIENTS SET name = %s WHERE Patient_ID = %s"
        cursor.execute(query, (new_name, patient_id ))
    elif choice == 2:
        new_age = int(input("Enter new age: "))
        query = "UPDATE PATIENTS SET age = %s WHERE Patient_ID = %s"
        cursor.execute(query, (new_age, patient_id))
    elif choice == 3:
        new_gender = input("Enter new gender: ")
        query = "UPDATE PATIENTS SET gender = %s WHERE Patient_ID = %s"
        cursor.execute(query, (new_gender, patient_id))
```

```

elif choice == 4:
    new_Phone = int(input("Enter new Phone number: "))
    query = "UPDATE PATIENTS SET Phone= %s WHERE Patient_ID = %s"
    cursor.execute(query, (new_Phone, patient_id))
else:
    print("Invalid choice.")
    return
    print("Patient details updated successfully!")

```

#program for delete patients

```

def delete_patients(cursor):
    patient_Id=input("enter patient's Id to delete:")
    query="DELETE FROM PATIENTS WHERE Patient_ID=%s"
    cursor.execute(query,(patient_Id,))
    print("Patient record deleted successfully!")

```

#program for add doctors

```

def add_doctors(cursor):
    name=input("Enter doctor's name:")
    specialization=input("Enter doctor's specialization:")
    phone=int(input("Enter doctor's phone number:"))
    email_id=input("Enter doctor's email_Id:")
    query="INSERT INTO
DOCTORS(Name,Specialization,Phone,Email)VALUES(%s,%s,%s,%s)"
    cursor.execute(query,(name,specialization,phone,email_id))
    print("Doctors added successfully!")

```

#program for view doctors

```
def view_doctors(cursor):
    cursor.execute("select * from DOCTORS")
    records=cursor.fetchall()
    print("Doctors records:")
    for record in records:
        print(record)
```

#program for update doctors

```
def update_doctors(cursor):
    doctor_id = int(input("Enter the Doctor ID to update: "))
    print("1. Update Name")
    print("2. Update specialization")
    print("3. Update email")
    print("4. Update Phone Number")
    choice = int(input("Choose the detail to update: "))
    if choice == 1:
        new_name = input("Enter new name: ")
        query = "UPDATE DOCTORS SET Name = %s WHERE Doctor_ID = %s"
        cursor.execute(query, (new_name,doctor_id))
    elif choice == 2:
        new_specialization= input("Enter new specialization: ")
        query = "UPDATE DOCTORS SET Specialization= %s WHERE Doctor_ID = %s"
        cursor.execute(query,(new_specialization,doctor_id))
    elif choice == 3:
        new_email = input("Enter new email: ")
```

```

query = "UPDATE DOCTORS SET Email = %s WHERE Doctor_ID =
%s"
cursor.execute(query, (new_email,doctor_id))
elif choice == 4:
new_Phone = int(input("Enter new Phone number: "))
query = "UPDATE DOCTORS SET Phone = %s WHERE Doctor_ID =
%s"
cursor.execute(query, (new_Phone,doctor_id))
else:
print("Invalid choice.")
return
print("Doctors details updated successfully!")

```

#program for delete doctors

```

def delete_doctor(cursor):
    doctor_id=input("enter doctor_id to be delete:")
    query="DELETE FROM DOCTORS WHERE Doctor_ID=%s"
    cursor.execute(query,(doctor_id,))

```

#program for schedule_appointment

```

def schedule_appointment(cursor):
    patient_id = int(input("Enter patient's ID: "))
    doctor_id = int(input("Enter doctor's ID: "))
    date = input("Enter appointment date (YYYY-MM-DD): ")
    time = input("Enter appointment time (hh:mm:ss): ")
    status = input("Enter appointment status (morning/afternoon): ")
    query = "INSERT INTO APPOINTMENTS (Patient_ID, Doctor_ID,
Date,Time,Status) VALUES (%s, %s, %s, %s,%s)"
    cursor.execute(query, (patient_id, doctor_id, date,time,status))
    print("Appointment scheduled successfully!")

```


#program for view appointments

```
def view_appointments(cursor):  
    cursor.execute("select * from APPOINTMENTS")  
    records=cursor.fetchall()  
    print("Appointments records:")  
    for record in records:  
        print(record)
```

#program for generate patients bills

```
def generate_bills(cursor):  
    patient_id=int(input("Enter patient's id :"))  
    amount=float(input("Enter bill amount:"))  
    payment_status=input("Enter payment status (Paid/Unpaid): ")  
    query="INSERT INTO  
BILLS(Patient_ID,Amount,Payment_Status)VALUES(%s,%s,%s)"  
    cursor.execute(query,(patient_id,amount,payment_status))
```

#program for view patients bill

```
def view_bill(cursor):  
    cursor.execute("select * from BILLS")  
    records=cursor.fetchall()  
    print(" Bill Records are:")  
    for record in records:  
        print(record)
```

#program for search patient by name

```
def search_patient(cursor):  
    name=input("enter patient's name to be search:")  
    query="SELECT * FROM PATIENTS WHERE Name=%s"  
    cursor.execute(query,(name,))  
    result = cursor.fetchall()  
  
    if result:  
        for row in result:  
            print(row)  
    else:  
        print("No patient found with that name.")
```

#program for search doctor by speciality

```
def search_doctor(cursor):  
    speciality=input("enter doctor's speciality to be search:")  
    query="SELECT * FROM DOCTORS WHERE Specialization=%s"  
    cursor.execute(query,(speciality,))  
    result = cursor.fetchall()  
    if result:  
        for row in result:  
            print(row)  
    else:  
        print("No doctor found with that speciality.")
```

#program for view medical history of patient

```
def patient_medical_history(cursor):
    patient_id = int(input("Enter Patient ID to view history: "))
    query = """
        SELECT          APPOINTMENTS.Date,          DOCTORS.Name,
DOCTORS.Specialization
        FROM APPOINTMENTS
        JOIN  DOCTORS  ON  APPOINTMENTS.Doctor_ID  =
DOCTORS.Doctor_ID
        WHERE APPOINTMENTS.Patient_ID = %s
    """

    cursor.execute(query, (patient_id,))
    records = cursor.fetchall()
    print("Medical History:")
    for record in records:
        print(record)
```

#main menu driven program

```
def main_menu():
    db = connect_to_db()
    cursor = db.cursor()

    while True:
        print("\n--- Hospital Management System ---")
        print("1.Add Patient")
        print("2.View Patients Records")
        print("3.Update Patients Records")
        print("4.Delete Patients Records ")
        print("5.Add Doctor")
        print("6.View Doctors Records")
        print("7.Update Doctors Records")
        print("8.Delete Doctors Records")
```

```
print("9.Schedule Appointments")
print("10.View Appointments")
print("11.Generate Patient's Bill")
print("12.View Patient's Bill")
print("13.Search patient by Name")
print("14.Search Doctor by speciality")
print("15.View Patient's Medical History")
print("16.Exit")
```

```
choice = int(input("Enter your choice: "))
```

```
if choice == 1:
    add_patient(cursor)
    db.commit()
elif choice == 2:
    view_patients(cursor)
elif choice == 3:
    update_patients(cursor)
    db.commit()
elif choice == 4:
    delete_patients(cursor)
    db.commit()
elif choice == 5:
    add_doctors(cursor)
    db.commit()
elif choice == 6:
    view_doctors(cursor)
elif choice == 7:
    update_doctors(cursor)
    db.commit()
elif choice == 8:
    delete_doctor(cursor)
```

```
db.commit()
elif choice == 9:
    schedule_appointment(cursor)
    db.commit()
elif choice == 10:
    view_appointments(cursor)
elif choice == 11:
    generate_bills(cursor)
    db.commit()
elif choice == 12:
    view_bill(cursor)
elif choice == 13:
    search_patient(cursor)
    db.commit()
elif choice == 14:
    search_doctor(cursor)
    db.commit()
elif choice == 15:
    patient_medical_history(cursor)
    db.commit()
elif choice == 16:
    print("Exiting program.")
    break
else:
    print("Invalid choice. Please try again.")

cursor.close()
db.close()
```

Run the program

```
if __name__ == "__main__":  
    main_menu()
```

OUTPUT

--- Hospital Management System ---

- 1.Add Patient
- 2.View Patients Records
- 3.Update Patients Records
- 4.Delete Patients Records
- 5.Add Doctor
- 6.View Doctors Records
- 7.Update Doctors Records
- 8.Delete Doctors Records
- 9.Schedule Appointments
- 10.View Appointments
- 11.Generate Patient's Bill
- 12.View Patient's Bill
- 13.Search patient by Name
- 14.Search Doctor by speciality
- 15.View Patient's Medical History
- 16.Exit

Enter your choice: 1

Enter patient's name: Reena

Enter patient's age: 23

Enter patient's gender: F

Enter phone number: 9817620187

Enter Admission date (YYYY-MM-DD): 2024-10-30

Patient added successfully!

Enter your choice: 2

Patients records:

(1, 'kanchan', 12, 'F', '9819872912', datetime.date(2024, 10, 10))

(2, 'Krishna', 20, 'M', '9028726399', datetime.date(2024, 10, 11))

(3, 'kajal', 34, 'F', '9986857968', datetime.date(2024, 10, 1))

(4, 'Reena', 23, 'F', '9817620187', datetime.date(2024, 10, 30))

(5, 'Vickey', 30, 'M', '9813248101', datetime.date(2024, 10, 23))

Enter your choice: 3

Enter the Patient ID to update: 2

1. Update Name

2. Update Age

3. Update Gender

4. Update Phone Number

Choose the detail to update: 1

Enter new name: Rohan

Patient details updated successfully!

Enter your choice: 4

Enter patient's Id to delete:2

Patient record deleted successfully!

Enter your choice: 5

Enter doctor's name:dr.Siddharth

Enter doctor's specialization:Otorhinolaryngology

Enter doctor's phone number:7296301732

Enter doctor's email_Id:siddharth1245@gmail.com

Doctors added successfully!

Enter your choice: 6

Doctors records:

(1,'dr.Rishi','Dermatology','9980789675','rishi100982@gmail.com')

(2,'dr.keshav','skinspecialist','3878508297','keshav-2038@gmail.com')

(3,'dr.Geetika','physiotherapist','7827291739','geetika1325@gmail.com')

(4,'dr.Naina','Pediatrics','1253790173','naina1234@gmail.com')

(5,'dr.Siddharth','Otorhinolaryngology','7296301732','siddharth1245@gmail.com')

Enter your choice: 7

Enter the Doctor ID to update: 4

1. Update Name

2. Update specialization

3. Update email

4. Update Phone Number

Choose the detail to update: 1

Enter new name: Vedant

Doctors details updated successfully!

Enter your choice: 8

enter doctor_id to be delete:5

Doctor record deleted successfully!

Enter your choice: 9

Enter patient's id:2

Enter doctor's id:2

Enter admission date (YYYY-MM-DD):2024-10-11

Enter appointment time (hh:mm:ss):03:30:00

Enter appointment status (morning/afternoon):afternoon

Appointment scheduled successfully!

Enter your choice: 10

Appointments records:

```
(1, 1, 1, datetime.date(2024, 10, 10),
datetime.timedelta(seconds=9000), 'afternoon')
(2, 1, 2, datetime.date(2024, 10, 10),
datetime.timedelta(seconds=15900), 'afternoon')
(3, 2, 2, datetime.date(2024, 10, 11),
datetime.timedelta(seconds=12600), 'afternoon')
(4, 3, 4, datetime.date(2014, 10, 30),
datetime.timedelta(seconds=39600), 'morning')
(5, 5, 4, datetime.date(2024, 10, 21),
datetime.timedelta(seconds=3600), 'afternoon')
```

Enter your choice: 11

Enter patient's id :4

Enter bill amount:400000

Enter payment status (Paid/Unpaid): Paid

Enter your choice: 12

Bill Records are:

```
(1, 1, 200000, 'unpaid')
(2, 3, 300000, 'Unpaid')
(3, 4, 400000, 'Paid')
(4, 4, 400000, 'Paid')
(5, 5, 540000, 'Paid')
```

Enter your choice: 13

Enter patients name to be search:kanchan

(1, 'kanchan', 12, 'F', '9819872912', datetime.date(2024, 10, 10))

Enter your choice: 14

Enter doctor's speciality to be search:Dermatology

(1, 'dr.Rishi', 'Dermatology', '9980789675', 'rishi19732@gmail.com')

Enter your choice: 15

Enter Patient ID to view history: 1

Medical History:

(datetime.date(2024, 10, 10), 'dr.Rishi', 'Dermatology')

(datetime.date(2024, 10, 10), 'dr.keshav', 'skin specialist')

Enter your choice: 16

Exiting program.

TABLES

DATABASE :- HOSPITAL_MANAGEMENT

Table 1:-PATIENTS

Table structure:-

DESC PATIENTS;

Field	Type	Null	Key	Default	Extra
Patient_ID	INT	NO	PRI	NULL	auto_increment
Name	VARCHAR(100)	NO		NULL	
Age	INT	YES		NULL	
Gender	CHAR(1)	YES		NULL	
Phone	VARCHAR(10)	YES		NULL	
Admission_date	DATE	YES		NULL	

Table Entries:-

SELECT * FROM PATIENTS;

Patient_ID	Name	Age	Gender	Phone	Admission_date
1	Kanchan	12	F	9819872912	2024-10-10
2	Krishna	20	M	9028726399	2024-10-11
3	Kajal	34	F	9986857968	2024-10-01
4	Reena	23	F	9817620187	2024-10-30
5	Vickey	30	M	9813248101	2024-10-23

Table 2:-DOCTORS

Table structure:-

DESC DOCTORS;

Field	Type	Null	Key	Default	Extra
Doctor_ID	INT	NO	PRI	NULL	auto_increment
Name	VARCHAR(100)	YES		NULL	
Specialization	VARCHAR(50)	YES		NULL	
Phone	VARCHAR(10)	YES		NULL	
Email	VARCHAR(100)	YES		NULL	

Table Entries:-

SELECT * FROM DOCTORS;

Doctor_ID	Name	Specialization	Phone	Email
1	dr. Rishi	Dermatology	9980789675	rishi1009@gmail.com
2	dr. Keshav	Skin Specialist	3878508297	keshav2038@gmail.com
3	dr. Geetika	Physiotherapist	7827291739	geetika1325@gmail.com
4	dr.Naina	Pediatrics	1253790173	naina1234@gmail.com
5	dr.Siddharth	Otorhinolaryngology	7296301732	ananya1265@gmail.com

Table 3:-APPOINTMENTS

Table structure:-

DESC APPOINTMENTS;

Field	Type	Null	Key	Default	Extra
Appointment_ID	INT	NO	PRI	NULL	auto_increment
Patient_ID	INT	YES	MUL	NULL	
Doctor_ID	INT	YES	MUL	NULL	
Date	DATE	YES		NULL	
Time	TIME	YES		NULL	
Status	VARCHAR(20)	YES		NULL	

Table Entries:-

SELECT * FROM APPOINTMENTS ;

Appointment_ID	Patient_ID	Doctor_ID	Date	Time	Status
1	1	1	2024-10-10	02:30:00	afternoon
2	1	2	2024-10-10	04:25:00	afternoon
3	2	2	2024-10-11	03:30:00	afternoon
4	4	3	2024-10-30	11:00:00	morning
5	5	5	2024-10-21	01:00:00	afternoon

Table 4:- BILLS

Table structure:-

DESC BILLS;

Field	Type	Null	Key	Default	Extra
Id	INT	NO	PRI	NULL	auto_increment
Patient_ID	INT	YES	MUL	NULL	
Amount	INT	YES		NULL	
Payment_status	VARCHAR(20)	YES		NULL	

Table Entries:-

SELECT * FROM BILLS;

Id	Patient_ID	Amount	Payment_status
1	1	200000	Unpaid
2	3	300000	Unpaid
3	4	400000	Paid
4	4	400000	Paid
5	5	540000	Paid